MUFFAKHAM JAH COLLEGE OF ENGINEERING & TECHNOLOGY

(SULTAN-UL-ULOOM EDUCATION SOCIETY)

COMPUTER SCIENCE AND ENGINEERING DEPARTMENT

SCIENTIFIC PROGRAMMING (ES 303CS)

QUESTION BANK 2024-2025

UNIT-1

Introduction to Scientific Programming and Python Basics

SHORT ANSWER TYPE QUESTIONS

1. What is scientific computing?
2. Define Python interpreter.
3. Mention any two key features of Python.
4. List any three numeric data types in Python.
5. What is indentation in Python?
6. Differentiate between global and local variables.
7. Explain the use of `range()` in Python with an example.
8. What are default arguments in Python functions?
9. Describe the role of comments in Python programming.
10. What is the purpose of the `import` statement in Python?

LONG ANSWER TYPE QUESTIONS

1. Write a Python program using if-elif-else to classify a number as positive, negative, or zero. Explain the output.
2. Demonstrate how to use a user-defined function with parameters and return type in Python.
3. Compare and contrast list, tuple, and dictionary with examples.
4. Analyze how control structures affect the flow of a Python program. Give examples using loops and conditionals.
5. Evaluate the advantages and limitations of using Python for scientific computing.
6. Discuss the significance of using libraries such as NumPy, Matplotlib, and SciPy in scientific programming.
7. Design a Python module containing at least one function and one variable.

Then, show how to import and use them in another script.

8. Create a recursive function in Python to compute the factorial of a number and explain the flow of execution.
9. Write a program in MATLAB to calculate and plot the sine function. Explain each command used.
10. Design a solution using nested loops in Python to display a multiplication table up to 10.

UNIT 2

Data Structures and Algorithms

SHORT ANSWER TYPE QUESTIONS

1. What is a list in Python?
2. How is a tuple different from a list in Python?
3. Define a dictionary with an example.
4. What is the purpose of the `set()` function in Python?
5. Name any two types of sorting algorithms.
6. Explain the use of the `.get()` method in dictionaries.
7. How does list slicing work?
8. What is meant by ―tuple unpacking‖?
9. How is a multidimensional array created in MATLAB?
10. Describe the difference between `.remove()` and `.discard()` in sets.

LONG ANSWER TYPE QUESTIONS

1. Write a Python program that demonstrates adding, removing, and modifying elements in a list.
2. Implement a program to search an element in a list using linear search and explain the steps.
3. Compare and contrast the characteristics of sets and dictionaries in Python with examples.
4. Analyze the difference between shallow copy and deep copy in Python lists.

5. Evaluate the advantages of using merge sort over bubble sort for large datasets.
6. Assess the usage of MATLAB's `zeros()`, `ones()`, and `eye()` functions in matrix creation.
7. Design a Python program that uses a dictionary to count the frequency of each word in a string.
8. Write a program to implement selection sort and display the sorted array.
9. Create a nested dictionary in Python and demonstrate how to access inner values.
10. Develop a MATLAB script to generate a 3D array using `rand()` and access a specific slice.

UNIT 3

Numerical Methods in Scientific Programming:

Short Answer Type Questions

1. What is the purpose of the bisection method in numerical computation?
2. Define truncation error.
3. State one advantage of the Newton-Raphson method.
4. What is a matrix in linear algebra?
5. Why does the bisection method require opposite signs at the interval ends?
6. What is the difference between forward and central difference methods?
7. Explain the term numerical integration in brief.
8. What is the role of eigenvalues in linear algebra?
9. Apply the Newton-Raphson method to find the root of $x^2 – 4 = 0$ with initial guess x0=1.
10. Use the trapezoidal rule to integrate $f(x) = x^2$ from 0 to 2 using 4 intervals.
11. Why might Newton-Raphson fail to converge?
12. Under what conditions does Simpson‘s rule provide better results than the trapezoidal rule?
13. When would you prefer Gauss-Seidel over Gaussian elimination?
14. Which method would you choose for a high-dimensional integral – Simpson‘s or Monte Carlo – and why?

Long Answer Type Questions

1. Explain the Bisection Method with its algorithm, flowchart, advantages, and disadvantages.
2. Compare the Bisection Method and Newton-Raphson Method in terms of convergence and applicability.
3. Write a Python or MATLAB program to implement the Trapezoidal Rule and demonstrate its use.
4. Implement the Bisection Method in Python to find a root of $f(x) = \cos(x) - x$, explaining each step.
5. Analyze the convergence behavior of Newton-Raphson and Bisection methods with examples.
6. Evaluate the accuracy of Trapezoidal and Simpson's Rule by comparing their error terms.
7. Assess the trade-offs between direct and iterative methods for solving linear systems.
8. Evaluate the limitations of numerical differentiation methods and suggest improvements.
9. Create a Python script that compares Trapezoidal, Simpson's, and Monte Carlo integration methods.
10. Design a MATLAB tool that numerically differentiates a function using forward, backward, and central difference methods with graphical output.

Unit 4

Data Analysis and Visualization

Short Answer Type Questions

1. What is data analysis?
2. Name any two libraries in Python used for data visualization.
3. List two types of data visualization techniques.
4. What function is used to load a CSV file in MATLAB?

5. Mention any two chart types supported by Seaborn.
6. Differentiate between a bar chart and a histogram.
7. What is the role of `groupby()` in data analysis using Pandas?
8. Why is Seaborn preferred over Matplotlib in some cases?
9. How do you filter rows in a Pandas DataFrame where the age is greater than 30?
10. Which Python function is used to plot a pie chart?
11. Identify a suitable visualization technique to show correlation between two numeric variables.
12. What are the key differences between Pandas DataFrame and MATLAB table for data manipulation?
13. Which platform—Python or MATLAB—would you choose for building real-time dashboards and why?
14. Evaluate the limitations of pie charts in data visualization.
15. Suggest a suitable visualization dashboard for a hospital‘s patient data monitoring system.
16. How would you design a visualization to represent customer feedback data across different regions?

Long Answer Type Questions

1. Describe the key steps involved in the data analysis process.
2. Explain different types of basic data visualization techniques with examples
3. Compare the visualization features of Python (Matplotlib/Seaborn) and MATLAB.
4. Explain the importance of data analysis in Python and MATLAB with suitable use cases.
5. Write a Python program using Matplotlib to draw a customized bar chart for —Total Bill vs. Day from the `tips.csv` dataset.
6. Demonstrate with code how to merge two dataframes in Pandas using a common key.
7. Analyze how `groupby()` and `splitapply()` work differently in Pandas and MATLAB with suitable examples.
8. Examine the effect of different parameters (color, marker, linestyle) in enhancing the clarity of a scatter plot.

9. Critically evaluate the use of Seaborn vs. Matplotlib in statistical data visualizations.
10. Assess the effectiveness of interactive dashboards in communicating insights to non-technical stakeholders.
11. Design a comparative dashboard using Matplotlib and Seaborn for sales data showing monthly trends, category performance, and regional distribution.
12. Create a complete data analysis pipeline in Python: Load data → Clean → Analyze → Visualize (use `tips.csv` or any dataset you prefer)

Unit-5

Emerging Technologies and Project Work

Short Answer Type Questions

1. What are the 3Vs of Big Data?
2. Define Machine Learning.
3. List any two components of IoT
4. Explain the importance of Scikit-learn in Machine Learning.
5. How does supervised learning differ from unsupervised learning?
6. How would you apply ML to predict heart disease using Scikit-learn?
7. Mention one real-time use of IoT in agriculture.
8. Differentiate between Edge computing and Cloud computing in IoT.
9. Compare the impact of Big Data in Civil and Mechanical Engineering.
10. What are the challenges in implementing IoT in smart cities?

Long Answer Type Questions

1. Describe the types of machine learning with examples.
2. Outline the steps involved in building a machine learning model using Scikit-learn.
3. Discuss the significance of IoT in the healthcare sector.
4. Explain the components and architecture layers of IoT.
5. Illustrate with examples how Big Data is used in predictive maintenance in mechanical engineering.

6. Apply ML and IoT to design a smart irrigation system.
7. Analyze the role of ML, IoT, and Big Data in developing smart cities.
8. Examine how Big Data analytics helps in supply chain optimization.
9. Critically evaluate the limitations of Scikit-learn for large-scale projects.
10. Assess the ethical issues in using IoT for personal finance management in banking.